

Bio-inspired navigation: Desert-ant inspired homing

Summary (~300 words):

Desert ants (*Cataglyphis* spp.) navigate in featureless environments by combining a path-integration strategy with

Algorithm (codified):

1. Outbound search:
 - Start at nest (0,0). Initialize heading.
 - Perform stochastic steps: each step of fixed length with occasional random turns.
 - Record each step vector in a list (for path integration).
 - Stop when the agent is within a small radius of the food.
2. Path Integration (PI):
 - Sum all outbound step vectors: $PI_vector = \text{sum of steps}$.
 - PI gives an estimate of the vector from nest to current position; negating it points towards nest.
3. Homing (fused cue):
 - Obtain a noisy sun-compass reading (absolute azimuth + Gaussian noise).
 - Form two direction estimates: PI-based angle and compass-based angle (to-nest estimate with noise).
 - Fuse angles by weighted average: $fused = w_PI * angle_PI + w_sun * angle_compass$.
 - Step towards fused direction until within nest radius or max steps.

Screenshot: trajectories and PI arrow

